
From AI Trustworthiness to Perceived User Trust in High-Risk AI: An Expert-Derived Conceptual Model

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Abstract: Artificial intelligence (AI) systems are increasingly used in high-risk domains where their decisions can significantly affect people’s lives. While trustworthy AI frameworks emphasize attributes such as accuracy, fairness, transparency, and reliability, these attributes do not necessarily translate into user trust. This study examines expert perspectives on this relationship through 14 interviews across four high-risk industries and six countries. Thematic analysis identified interconnected technical, human, organizational, and contextual factors that experts believe influence perceived trust. Participants associated trust with system performance and reliability, but also emphasized usefulness, contextual relevance, transparency, accountability, human agency, user control, and organizational reputation. Building on these findings, we propose an expert-derived conceptual framework that extends the established distinction between AI trustworthiness and human trust by explaining how the two may become connected. Rather than assuming that trustworthy system characteristics directly generate trust, our framework proposes that trustworthiness attributes are translated into perceived trust through users’ interpretation and experience of the system. This interpretive stage is shaped by two distinct sets of factors: user dispositional factors, such as demographics, risk tolerance, AI literacy, prior AI experience, and propensity to trust, and contextual influences, such as organizational reputation, social and domain norms, and regulatory context. Perceived trust in turn influences reliance and system use, while remaining distinct from them; because reliance is directly observable, experts often infer trust from it, even though continued use is an ambiguous signal that may instead reflect convenience, necessity, or a lack of alternatives. This framework represents experts’ perspectives rather than direct evidence of users’ trust experiences and provides a foundation for future empirical research with end-users.

Keywords: Trustworthiness in High-risk AI Systems, Gap between Trustworthiness and Perceived Human Trust, Human-AI Trust, Geopolitical Impact of AI Regulations on AI Trustworthiness Factors, AI in Healthcare, AI in Human Resources Technologies, Responsible AI Innovations.

1. INTRODUCTION

In the context of Artificial Intelligence (AI), trust and trustworthiness are critical for the successful societal adoption of AI systems [3]. Trust and trustworthiness are distinct yet often mistaken as interchangeable concepts [20]. It is possible to build an AI system that meets all established trustworthiness criteria such as transparency, robustness, and fairness yet still be rejected by users.

Trust in AI refers to a human’s psychological willingness to rely on an AI system despite uncertainty or potential risk, based on the belief that the system will act in their best interest and perform as expected [11]. While trustworthiness refers to the qualities or attributes of the AI system itself that justify trust, such as reliability, accuracy, fairness, robustness, transparency, and safety [15]. Experts highlighted a gap between

the factors that define trustworthy AI and those that influence user trust. For example, explainability does not always increase trust in practice [12]. Poorly designed explanations can confuse users or reduce trust [2, 7]. Therefore, technical features intended to improve trustworthiness do not always translate into greater user trust.

We suspect that one of the core reasons people distrust AI systems that claim to assess trustworthiness is that these systems are often developed in silos. There is growing evidence of user dissatisfaction, mistrust, and repeated cases of harmful or inappropriate AI behavior [9]. A key issue is that people are rarely asked how they want to be treated by AI systems or which values and concerns matter most to them [4]. As a result, definitions of trustworthy AI are frequently shaped within isolated legal or technical communities rather than being grounded in users’ lived experiences and expectations.

At the same time, we still lack a clear understanding of how to measure trust in AI systems because the behavioral signals of trust remain ambiguous [17]. For example, high user retention is often treated as evidence of trust, but experts suggest this may be misleading. Users may continue using AI due to a lack of alternatives, practical value, or ease of use. As a result, usage metrics alone may not capture important emotional, social, and contextual factors that influence trust. We conducted interviews with 14 experts working across high-risk AI industries, as defined by the EU AI Act [10]. Our interview questions were grouped into two categories: trustworthiness in AI systems and experts’ views on user trust. The first explored how trustworthy AI systems are designed, developed, assessed, and validated, while the second examined how experts conceptualize end-users perceived trust, including their needs, expectations, and vulnerabilities. These insights can inform hypotheses for future research on the relationship between AI trustworthiness and user trust.

2. PRIOR RESEARCH ON TRUST AND TRUSTWORTHINESS

Recent research highlights an important gap between AI trustworthiness and user trust, particularly in high-risk domains. Trustworthiness is typically assessed through technical factors such as transparency, fairness, robustness, privacy, accountability, and safety [14]. While these approaches help evaluate whether an AI system satisfies technical and regulatory requirements, they do not fully explain whether users actually trust such systems in practice. This creates a gap between objective system trustworthiness and subjective user trust perception.

Several studies show that perceived trust is shaped by contextual and psychological factors beyond technical reliability. Transparency alone does not guarantee trust, especially in high-risk settings where reliability and accountability may be more influential [6]. In healthcare, physicians’ trust in AI-based clinical decision support systems is affected not only by explainability, but also by concerns about professional identity, workflow integration, and accountability [1]. Similarly, technical measures alone are insufficient to create trust, particularly when AI systems are too opaque for users to evaluate independently [5]. Therefore, even technically trustworthy systems may fail to gain user trust if social, organizational, and contextual concerns are not addressed. For instance, in high-risk domains, public trust is strongly influenced by confidence in the organizations behind AI systems [18]. The concept of vigilant trust captures this more balanced form of trust where users neither blindly trust nor reject AI. Instead, they remain aware of potential risks while continuing to use AI when its benefits, safeguards, and accountability justify doing so [8].

The relationship between trustworthiness and perceived trust is therefore highly complex. Perceived trust cannot be predicted solely through technical trustworthiness metrics. Diverse user groups interpret trust, fairness, and safety in generative AI conversational systems differently depending on their personal attitudes and experiences [23]. Perceived trust is dynamic and shaped by users’ ongoing assessment of risk and uncertainty. For instance, users’ reliance on AI-assisted advice changes according to task risk and usability. Users may rely on AI even when trust is limited, or distrust AI despite strong technical performance [22].

3. METHODS

As shown in Table B-1, the group of experts we interviewed represents a diverse sample spanning North America, Europe, Asia, and Australia, though the majority are based in Europe. Participants represented senior positions across industry, academia, and healthcare, including CEOs, founders, professors, chief data

scientists, and research leaders. Most participants were from the healthcare sector, followed by human resources technology, with additional representation from education and e-commerce. Ethical approval was obtained from the University of Nottingham School of Computer Science Research Ethics Committee (REC) (Approval ID: [CS-2025-SA3]), and all participants provided informed consent prior to participation. We used descriptive coding and thematic analysis to identify patterns and themes in the interview data. Codes were grouped into broader categories based on their underlying meaning, allowing key issues and insights to emerge inductively from the data [13].

4. FINDINGS: SYSTEM DESIGN

4.1. AI System Purpose and Functionality Overview (Q2)

Experts discussed one or two AI systems they had helped develop, including their purpose, functionality, users, and deployment context. Table B-2 summarizes these AI use cases. 23 AI systems ranged from HR tools and clinical decision support to generative assistants and robotics. Most systems supported human decision-making within existing workflows rather than replacing users. The use cases were grouped into six categories: Agentic AI, AI Governance, Clinical AI, Enterprise AI, HR AI, and Personalization/Behavioral AI. These categories highlight common design and evaluation practices and the contextual factors currently shaping AI development and deployment.

4.2. Responsible AI Standards and Ethical Practices (Q3)

As illustrated in Figure D-1, the findings from questions 3 (Q3) show that responsible AI is approached as a multi-layered framework rather than a single set of practices. Expert responses cluster around four main themes: Custom Guidelines and Frameworks, AI Ethics, Regulations, and AI Safety and Security.

Table C-1 highlights that Custom Guidelines and Frameworks (9 sources, 15 references) and AI Ethics (7 sources, 11 references) were the most prominent governance approaches, supported by external standards such as ISO, OECD, and UNESCO. Participants associated trustworthy AI with reliability, transparency, usability, and efficiency, while emphasizing legal compliance through Regulations (6 sources, 10 references), particularly GDPR and the EU AI Act, for data privacy and impact assessments. AI Safety and Security (4 sources, 4 references) was also identified as important for technical robustness and cybersecurity.

Experts further distinguished responsible AI from compliance-driven approaches, emphasizing that meeting regulatory requirements does not necessarily ensure ethical or safe outcomes. Responsible AI was understood to encompass broader ethical considerations that may extend beyond existing regulations, particularly where requirements lack operational clarity. As P10 explained: “Something that I usually say is that not necessarily because you’re compliant, you’re ethical. for me, there is a gap between what can be ethical, what can be safe and what can be compliant.” Participants also noted a gap between evolving ethical expectations and slower regulatory development, with leading organizations working to bridge that gap.

4.3. Challenges in Implementing Responsible AI (RAI) Practices (Q4)

Interview data identified several challenges to implementing responsible AI practices. Participants highlighted a combination of technical, organizational, and regulatory barriers that complicate the adoption and operationalization of responsible AI as illustrated in Figure D-2 and listed below:

- Insufficient technical capability and regulations to manage AI harms
- EU Restrictions
- Misalignment between AI regulation, ethics, and organizational practice
- Organizational barriers to AI implementation
- Healthcare-specific challenges in AI implementation
- Uncertainty in evaluating long-term AI impacts
- Lack of clearly defined quantitative metrics and acceptable thresholds for evaluating AI trustworthiness.

As shown in Table C-2, the most prominent challenge is the limited understanding of current AI capabilities and emerging behaviours. AI can infer information beyond what is explicitly provided, creating downstream risks such as bias, privacy concerns, and exposure of sensitive information. Despite progress in measuring bias in AI systems [16, 21], current methods often fail to capture real-world complexities, particularly in deployed systems and high-risk contexts. More robust, context-sensitive, and practical approaches to evaluating bias and fairness are still needed [7, 19]. Similarly, participants highlighted a lack of clearly defined quantitative metrics and acceptable thresholds for assessing broader trustworthiness dimensions, including unstructured data quality, bias, fairness, and related factors. This makes it difficult for organizations to determine whether an AI system is sufficiently trustworthy in practice, especially when trustworthiness depends on contextual, domain-specific, and evolving criteria rather than a single universal benchmark.

Participants also highlighted the lack of universally adopted AI governance regulations and the ongoing challenge of keeping safeguards aligned with rapidly evolving technologies. They also noted that defining ethical AI remains difficult, as ethical principles vary across cultural, organizational, and contextual settings, making universal standards hard to establish.

Another key challenge is the misalignment between regulation, ethics, and organizational practice. New AI regulations often lack clear operational and implementation guidance, leaving organizations struggling to translate high-level principles such as those from UNESCO Recommendation on the Ethics of Artificial Intelligence [24] into concrete features, testable systems, and safety mechanisms. This creates ongoing tension, particularly in areas affecting human rights, where a purely compliance-driven approach may be insufficient. As P10 explained, “you’re in constant negotiation around topics that have to do with affecting human rights.” Effective implementation therefore requires an innovation mindset, positioning responsible AI as a driver of system improvement rather than a checklist for satisfying legal requirements. This involves creating opportunities for critical reflection, as P10 emphasized: “But AI ethics is meant to question.” However, embedding this work within organizational structures remains challenging. The participant described the need for “a space where you can contest and you can question,” while noting that “you don’t know where you should be integrated in an organization.”

In Europe, AI and data regulations are generally stricter, but the regulatory landscape remains fragmented. Frameworks such as GDPR and the EU AI Act provide high-level requirements with limited implementation guidance, creating gaps between legal compliance and ethical expectations. Participants cited challenges such as restrictions on US-based cloud services and stricter compliance requirements for public institutions than for private organizations. Additionally, organizational and domain-specific constraints complicate responsible AI implementation. Organizations face limited resources, competing priorities, and pressure to demonstrate business value, while highly regulated sectors such as healthcare must navigate sensitive data, regulatory requirements, and concerns about sharing data with large language models.

Human factors and organizational culture also emerged as barriers to responsible AI implementation, particularly when translating ethical principles into practice while balancing competing priorities. As P5 explained, “how it happens in practice, I think it’s far more embedded in the institution’s priorities, which are of course not just about ethics principles, they’re about finances.” The participant further described implementation as an “ongoing negotiation of institutional priorities and researcher priorities.” Embedding responsible AI requires not only ethical guidance but also organizational support to reconcile these principles with operational and financial demands.

5. FINDINGS: SYSTEM EVALUATION

The previous section examined the AI systems discussed by experts and the responsible AI standards used in their design. The current section explores how experts assess trustworthiness and the behavioral signals and external indicators to evaluate user trust and confidence.

5.1. User Feedback Collection and Analysis Methods (Q5)

As shown in Figure D-3, participants described a multi-layered feedback approach that combines direct user input, behavioral data, and structured evaluations rather than relying on a single feedback channel. The findings suggest that effective feedback ecosystems are continuous and integrated into user-centered design

to capture both user experience and system performance.

Table C-3 illustrates the most prominent theme was user-centered design inputs (10 sources, 18 mentions), emphasizing direct user engagement through interviews, surveys, usability testing, feedback meetings, user research, and support channels. Explicit feedback mechanisms (4 sources, 8 mentions) and complementary system and user logs (4 sources, 4 mentions) further supported continuous feedback and real-world usage monitoring. Participants also highlighted evaluation and testing approaches (6 sources, 7 mentions), including benchmarking against performance standards, continuous or “instant” learning loops, and incident reporting. In regulated and high-risk domains such as healthcare, adverse event reporting was identified as critical for detecting failures and unintended consequences. Some participants described more structured feedback aggregation methods, including weighted voting systems that consider both user preferences and the strength of user sentiment which helps with producing more balanced and representative outcomes.

5.2. Evaluation of Ethical Policy Effectiveness (Q6)

The findings highlight the importance of organizational readiness, implementation tools, and evaluation mechanisms. Organizational readiness (3 sources, 4 mentions) and evaluation methods (4 sources, 5 mentions) were the most frequently cited approaches for assessing trustworthiness and improving AI performance as shown in Table C-4. Organizations need robust evaluation mechanisms to conduct studies, interview the users affected by AI systems, and subject their practices to both internal and external audits to ensure “responsible AI” is continuously tested against real-world outcomes rather than assumed. Second, they need concrete implementation tools, such as operational frameworks and requirements checklists, that translate abstract regulatory and ethical principles into repeatable, measurable actions teams can apply and improve over time. Underpinning the other two, they need organizational readiness: a culture and structure capable of sustaining responsible AI. This includes building AI literacy across the institution so teams can recognize and reason about ethical issues, pursuing company-level compliance certifications that hold the organization to external standards, and establishing a multidisciplinary AI ethics governance committee to provide oversight and accountability.

5.3. User Trust Measurement and Monitoring (Q7)

Question 7 explored how developers perceive, measure, and monitor user trust. As shown in Table C-5, participants (11 sources, 30 mentions) described three main approaches: behavioral observation, self-reported perceptions, and structured trust measurement frameworks. Behavioral observation focused on usage patterns, adoption rates, system logs, and interactions as indicators of engagement and acceptance. However, participants cautioned that usage does not necessarily equate to trust, as users may continue using AI systems due to convenience, organizational requirements, or lack of alternatives despite concerns about reliability or risk. Self-reported measures included direct user feedback, interviews, and explicit trust-related questions, providing insight into attitudes, perceptions, and concerns that may not be visible through behavioral data alone.

Participants also emphasized that trust is dynamic and context-dependent, requiring multidimensional measurement frameworks that integrate behavioral and self-reported data and account for the interdependent relationships among the AI system, the individual, the use context, society, and the broader environment. Examples included trust thresholds, surveys, and frameworks assessing dimensions such as digital trust, human-AI role alignment, outcome accuracy, explainability, and contextual fit score.

6. FINDINGS: USER CHARACTERISTICS AND IMPACT

The findings from Q8–10 explored experts’ perspectives on how users experience, assess, and develop trust in AI systems in practice.

6.1. User Base Characteristics and Usage Scale (Q8)

As shown in Table C-6, AI systems mentioned during our interviews are deployed across a wide range of contexts, from large-scale public platforms with over 72,000 developers and 2.63 million monthly users to

organizational deployments serving around 4,000 users, and specialized clinical networks involving approximately 16,296 professionals and 8,000 patients. Participants represented diverse sectors, including healthcare, human resources, academia, governance, data science, and software development, bringing expertise in AI implementation, validation, regulation, and operational use.

Despite differences in scale and application, the data highlights the multidisciplinary nature of AI adoption, involving technical experts, clinicians, managers, researchers, and diverse user groups.

6.2. User Trust Challenges in AI Systems (Q9)

As shown in Figure D-4, seven key emerging themes highlight multi-dimensional barriers to trust AI systems with the most prominent concerns relating to data privacy, governance, and system reliability. Challenges to user trust across 6 sources and 27 references was mentioned in Table C-7. Trust declined when users lacked confidence in how personal data was collected, shared, or used, especially in healthcare contexts where trust depends on responsible data handling, transparency, accountability, and security. Participants also highlighted governance and accountability concerns, including uncertainty about responsibility for AI errors and inconsistent validation practices, particularly in healthcare.

Performance and contextual relevance also emerged as factors shaping perceived user trust. Participants described concerns about inaccurate outputs, hallucinations, and response speed. As P12 noted, “when it comes to hallucinations, can we trust the output? [...] we’ve got that again and again.” The same participant emphasized expectations for responsiveness: “People most of the time say it has to be fast, has to be effective. Yeah, no latency.” Beyond accuracy and speed, participants highlighted the importance of tailoring responses to users’ circumstances. Reflecting on an HR chatbot’s limitations, P11 explained, “that was a challenge because we couldn’t then personalize,” linking this limitation to “trust in the usefulness of results.” Access to technical support was also described as important, with P2 noting that “everyone should be able to seek support and create tickets” through a help desk. Together, these accounts highlight reliability, responsiveness, and contextual relevance as considerations in perceived trust, alongside the availability of support when users encounter difficulties.

6.3. Features Enhancing User Trust and Confidence (Q10)

Trust in AI emerged as a concept shaped by several interrelated factors as illustrated in Table C-8. The most prominent theme was System Performance & Usefulness, with participants emphasizing that trust develops when AI systems are accurate, reliable, practical, and capable of improving decision-making. Trust was further strengthened by traceable outputs, consistent performance, and personalized, context-specific support, particularly in healthcare.

Participants also identified Governance, Accountability & Ethical Design, Transparency, and Human Agency & Control as critical foundations of trust. Trustworthy AI was viewed as something embedded throughout the AI lifecycle, supported by fairness, bias mitigation, explainability, accountability, and mechanisms that allow users to understand, monitor, challenge, and override AI recommendations. Human oversight remained essential to preserving user agency and control. Additional factors included Data Privacy, Organizational Reputation, and User Empowerment. Participants emphasized privacy-preserving designs, user control over data, and confidence in the organizations developing AI systems. They also stressed the importance of involving users in defining AI use cases and supporting AI literacy, arguing that trust is more likely to develop when users help shape solutions and understand both the capabilities and limitations of AI.

7. EXPERT-DERIVED CONCEPTUAL MODEL OF AI TRUST-WORTHINESS AND PERCEIVED USER TRUST

Drawing together the interview themes, we propose an expert-derived conceptual model of the relationship between AI trustworthiness and perceived user trust, as demonstrated in Figure 1. Our framework was developed by synthesizing a recurring pattern across the expert interviews: while trustworthiness is essential, it does not automatically translate into perceived user trust. Although trustworthiness and trust are distinct concepts, our findings suggest that they are interconnected through users’ interpretation and experience of AI

systems. Experts first described trustworthy AI through system-level qualities such as accuracy, reliability, fairness, bias mitigation, explainability, accountability, privacy, and monitoring. These qualities form the foundation of our framework because they indicate whether an AI system may be considered worthy of trust. However, our findings also suggest that users do not experience trustworthiness as a checklist of technical or governance attributes. Rather, users encounter AI systems within specific workflows, decision contexts, and organizational environments.

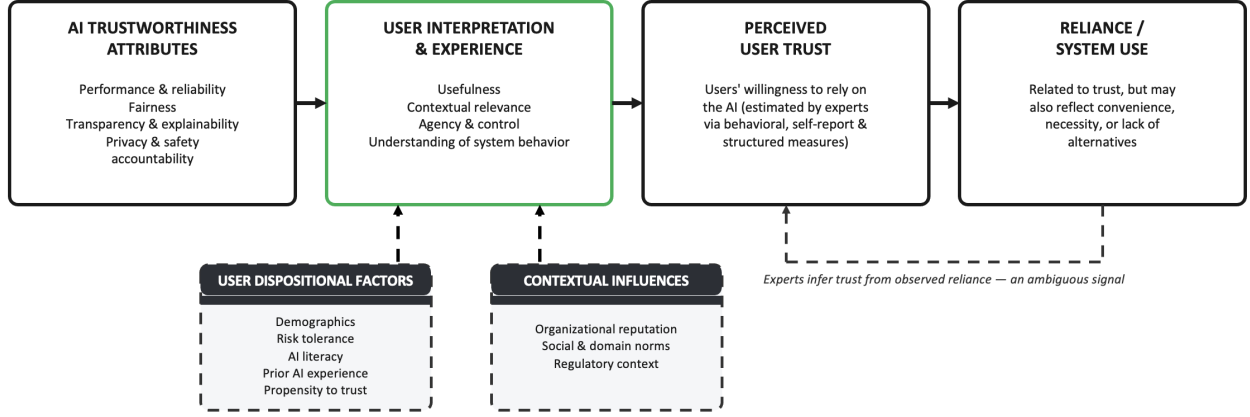


Figure 1. Expert-derived model linking trustworthiness attributes, user interpretation and experience, perceived trust, and reliance. Interpretation is shaped by user dispositional and contextual factors; experts partly infer trust from observed reliance.

For trustworthiness attributes to become meaningful to users, the system must demonstrate practical usefulness, contextual relevance, understandable behavior, and sufficient opportunities for users to question, challenge, or override AI outputs. Accordingly, the central stage of our framework is user interpretation and experience, including usefulness, contextual fit, agency and control, and understanding of system behavior. Our framework proposes that this interpretive stage is shaped by two distinct sets of factors. The first is user dispositional factors, such as demographics, risk tolerance, AI literacy, prior AI experience, and general propensity to trust: individual differences that users bring with them and that shape how the same trustworthiness attributes are read and weighed. Unlike the other components of the framework, these dispositional factors are drawn from the existing literature [17] rather than from our expert interviews. The second is contextual influences, such as organizational reputation, social and domain norms, and regulatory context: environmental conditions surrounding the deployment of the system rather than properties of the individual user, which experts raised directly in their accounts. Both sets of factors feed into how trustworthiness attributes are interpreted and experienced.

Perceived user trust, in turn, may lead to reliance and system use. Our framework treats these as related but distinct: continued use may reflect convenience, organizational requirements, practical value, or a lack of alternatives rather than genuine trust. At the same time, because reliance and system use are directly observable while trust is not, experts frequently work backwards from behavior, inferring users’ trust from adoption rates, usage patterns, and system logs. We represent this in the framework as a feedback link from reliance and system use to perceived user trust, capturing how experts estimate trust in practice. This link is deliberately marked as an ambiguous signal: as participants cautioned, usage does not necessarily equate to trust, so inferring trust from observed reliance can misrepresent whether users actually trust a system.

The proposed framework should therefore not be read as a simple causal pathway in which trustworthiness automatically produces trust. Rather, it represents an expert-derived synthesis of how trustworthiness attributes may become relevant to perceived user trust once they are filtered through an interpretive stage that is jointly shaped by who the user is and the context in which they encounter the system, and of how experts, in turn, infer that trust from observable reliance.

The framework aligns with existing literature that distinguishes system-level trustworthiness from human trust and recognizes trust as context-dependent rather than an inherent property of an AI system. It adds to this literature by integrating these perspectives into a multi-level pathway that explains how trustworthiness attributes may be translated, through an interpretive stage shaped jointly by user dispositional factors

and contextual influences, into perceived trust and, subsequently, reliance, while also capturing how experts infer trust from that reliance. In doing so, the framework provides a structure for examining experienced trustworthiness as an intermediate mechanism, for separating perceived trust from behavioral reliance, and for making explicit the inferential step through which observed reliance is read as evidence of trust. Future research can empirically test these proposed relationships using end-user studies that operationalize user dispositional factors (e.g., demographics, risk tolerance, AI literacy) and contextual influences (e.g., organizational reputation, regulation) as distinct, measurable inputs to the interpretation stage, and that examine how closely expert inferences drawn from reliance track users' self-reported trust, across different domains, risk levels, user groups, and organizational contexts. Our framework can be further validated and consolidated by operationalizing its constructs into measurable variables and testing the strength and direction of the proposed relationships using qualitative, quantitative, and longitudinal studies.

8. CONCLUSION

This study identified a gap between how organizations establish AI trustworthiness and what experts believe shapes user trust in high-risk domains. Interviews with 14 AI experts suggested that organizations primarily approach trustworthiness through governance, compliance, and technical safeguards. However, experts emphasized that perceived user trust also depends on practical value, user experience, accountability, user control, and context. Technical performance alone therefore cannot establish whether users trust a system. Similarly, adoption or continued use may reflect convenience, necessity, or a lack of alternatives rather than trust.

Based on these findings, we propose a conceptual framework that explains how AI trustworthiness may contribute to perceived user trust. Its central contribution is the concept of experienced trustworthiness: how users encounter, interpret, and judge a system's technical and governance attributes through their interactions with it. The framework positions user interpretation and experience as the stage through which trustworthiness is translated into perceived trust and identifies two distinct sets of factors that shape this stage; user dispositional factors such as demographics, risk tolerance, and AI literacy and contextual influences such as organizational reputation, social norms, and regulation. This helps explain why safeguards may not build trust unless users can recognize and understand their value, and why the same system may be trusted differently by different users or in different settings. The framework also distinguishes trust from reliance, recognizing that using an AI system does not necessarily mean trusting it, while making explicit that experts often infer trust from observed reliance an inferential step that can be misleading precisely because usage is an ambiguous signal.

The framework reflects expert perspectives rather than direct evidence of users' experiences. Future research should test and refine the proposed relationships with end users and develop measures that distinguish system trustworthiness, experienced trustworthiness, perceived trust, and appropriate reliance. This work should also examine the influence of user dispositional factors and contextual conditions and assess whether expert inferences of trust drawn from reliance align with users' own reported trust.

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Appendix A: Interview Framework and Question Set

Expert Background

- What is your current job role and main responsibilities?

System Design

To examine how organizations design and develop trustworthy AI systems, we asked participants the following questions:

- Can you describe what your system does? If possible, please share a website link where I can learn more.
- Can you tell me about the best practices and ethical standards your organization follows to ensure AI is implemented responsibly?
- What challenges do you face meeting these practices? Can you give me some examples?

System Evaluation

To understand how organizations evaluate the trustworthiness of AI systems in practice, participants were asked the following questions:

- How do you collect user feedback?
- Do you assess the effectiveness of your ethical policies and best practices? If yes, how?
- Do you track user trust? If yes, how?

Perceived Trust and User Understanding

To investigate how experts conceptualize end-users, including their needs, expectations, and vulnerabilities, participants were asked the following questions:

- Could you share some insights about your user base — for example, the types of users you serve, their backgrounds, and the overall scale of usage?
- Can you identify some user challenges related to trust in AI systems?
- Are there particular features or approaches you’ve found that make users feel more confident or trusting when engaging with your AI systems, such as explainability or transparency?

Open-Ended Feedback

- Any Questions or Further Comments?

Appendix B: Interview Participant Demographics and Use Cases

ID	Gender	Role	Industry	Country	Region
P1	F	UX Researcher	Healthcare	USA	NA
P2	M	Senior Scientist / Deputy Lab Lead	Healthcare	Belgium	EU
P3	F	AI Expert	Healthcare	USA	NA
P4	M	Director of AI Compliance	Healthcare	Sweden	EU
P5	F	Professor	Healthcare	USA	NA
P6	M	Associate Professor	Healthcare	UK	EU
P7	F	VP of AI - CEO	HR Tech	Australia	Au
P8	M	Chief Data Scientist	HR Tech	Singapore	Asia
P9	M	CEO and Founder	HR Tech	Germany	EU
P10	F	Senior Responsible AI Advisor	HR Tech	Germany	EU
P11	M	Head of Continuous Improvement	HR Tech	UK	EU
P12	M	Responsible AI Researcher	Education	Germany	EU
P13	M	CEO and Founder	HR Tech	UK	EU
P14	M	Director of Data Analytics	e-Commerce	USA	NA

Table B-1. List of interviewees, along with their gender, role, industry, country, and region. The names have been pseudonymized and replaced with identifiers P1 through P14.

P Number	Use cases discussed by experts	Question / Section
P1	Internal cloud tools for distributing ML/AI resources across engineers	System design / user journey
P1	Health monitoring app for elderly (sensor-based alerts)	Trust features / past experience
P2	Agentic interface for public service form filling	System design
P2	Insurance/product discovery via AI assistant	User base discussion
P3	Document AI for extracting structured data (HR/business docs)	System design
P3	LLM benchmarking & evaluation systems	Evaluation discussion
P4	HR skills prediction system (workforce analytics)	System design / journey
P5	Clinical decision support for organ transplant decisions	System design
P5	Fall prevention AI using radar sensors	Implementation example
P6	Clinical trial data summarization (AI-generated patient narratives)	System design
P7	Brain simulation / digital twin AI	System design
P7	AI for epilepsy surgery planning	Clinical use case
P7	Deep brain stimulation optimization	Clinical use case
P8	Responsible AI learning platform (Horizon app, CO ₂ tracking, safety filters)	System design
P9	Data quality ML system for anomaly detection in pipelines	System design
P10	AI platform for executive coaching & session analysis	System design
P10	Clinical safety AI detecting psychological risks	High-risk AI discussion
P11	HR chatbot ("Vera") for family policy queries	System design
P12	Governance of healthcare AI tools (diagnostic + admin systems)	Governance discussion
P13	Data platform for AI lifecycle governance (model benchmarking + risk assessment across multiple AI systems)	System design / governance
P13	AI benchmarking using risk categories (e.g., safety, societal risk, regulated advice)	Evaluation / benchmarking
P14	Customer experience AI systems (recommendation/personalization systems with bias/manipulation risks)	System design context
P14	HR AI systems lifecycle (design, evaluation, and user interaction analysis)	System design / evaluation

Table B-2. AI use cases discussed throughout the interview process.

Appendix C: Thematic Analysis Coding Tables

Name	Sources	References
[Q3] Responsible AI Standards and Ethical Practices	13	41
[Theme] AI Ethics	7	11
Efficiency	1	1
RAI Toolkits	1	1
Reliability	2	2
Transparency in system limitations	1	1
Trustworthy AI	1	2
Usability	1	1
[Theme] AI Safety and Security	4	4
AI Safety	1	1
Cybersecurity Standards	3	3
[Theme] Custom Guidelines and Frameworks	9	15
Internal Policies	1	1
Internal Standards	2	2
ISO Standards	1	1
National Standards	3	3
OECD Guidelines	1	1
UNESCO Principles	1	1
[Theme] Regulations	6	10
Device Regulations	2	3
EU AI ACt	3	3
GDPR	5	7
Data Privacy	1	1
DPIA	1	1

Table C-1. Coding summary for Responsible AI Standards and Ethical Practices (Q3).

Name	Sources	References
[Q4] Challenges in Implementing Responsible AI Practices	13	41
[Theme] Controls and Balance	1	1
Keeping the balance between safety and censorship	1	1
[Theme] EU Restrictions	3	6
More strict regulations when changing the setting for the same device	1	1
Public institutions are more strict in implementing regulations than private ones	1	1
US Cloud Services not Compliant in EU	2	4
[Theme] Healthcare-specific challenges in AI implementation	2	3
Health Data is Personal	1	1
Not accurate AI data for FDA clinical trials	1	1
Sharing health data with LLM's	1	1
[Theme] Insufficient technical capability and Regulations to manage AI Harms	7	13
AI Inferring PII or Sensitive info from text	2	2
Challenges in Measuring bias	1	2
Emerging unprecedented AI Challenges	1	1
Hard to control and limit the use of AI to company policies	1	1
Hard to implement AI Guardrails	1	1
Human-in-the-loop for Agentic AI	1	1
Inventing Novel Methods of Application-testing for bias	1	1
Novelty of the issues and lack of globally approved guidelines	2	2
Uncertain and biased opinions about what is ethical	1	2
[Theme] Misalignment between AI regulation, ethics, and organizational practice	2	5
Gap between Regulations and ethics and safety	1	1
Gaps in operational and implementation guidance within new AI regulations	1	2
RAI as innovation practice and not a compliance practice	1	2
[Theme] Organizational barriers to AI implementation	4	5
Building Trust in the AI System	1	1
Institutions competing priorities and finances	1	1
Not user-friendly or efficient enough	1	1
Resource Constraints in RAI	1	1
Should prove the AI's value-add	1	1
[Theme] Uncertainty in evaluating long-term AI impacts	2	3
AI Impact Assessment	1	2
Downward Negotiation of Harm	1	1

Table C-2. Coding summary for challenges in implementing responsible AI practices (Q4).

Name	Sources	References
[Q5] User Feedback Collection and Analysis Methods	12	37
[Theme] Data Collection Methods	7	12
Explicit feedback collected by software	4	8
System & User Logs	4	4
[Theme] Evaluation & Testing Approaches	6	7
Adverse event reporting	1	1
Benchmarking	2	2
Instant Learning	2	2
Scientific Publications & Articles	1	2
[Theme] User-Centered Design Inputs	10	18
JTBD & Tools	1	1
Quadratic Voting System to collect user feedback	1	1
Regular Feedback Meeting	2	2
Running Surveys	3	3
Support emails and tickets	1	1
Usability Testing	3	3
User Interviews	5	5
User Requirement Research	2	2

Table C-3. Frequency of user feedback collection and analysis methods discussed by participants (Q5).

Name	Sources	References
[Q6] Evaluation of Ethical Policy Effectiveness	6	11
[Theme] Evaluation Mechanisms	4	5
Conduct Studies & Interview Users	3	4
Internal & External Audits	1	1
[Theme] Implementation Tools	2	2
Operational Frameworks	1	1
Requirements Checklist	1	1
[Theme] Organizational Readiness	3	4
Company-level Compliance Certifications	1	2
Develop AI Literacy in the institution	1	1
Multidisciplinary AI Ethics Governance Committee	1	1

Table C-4. Frequency of themes related to evaluating the effectiveness of ethical AI policies and practices (Q6).

Name	Sources	References
[Q7] User Trust Measurement and Monitoring	11	30
[Theme] Behavioral and Observational Trust Signals	5	7
Adoption Rate	5	5
Defensive Information Seeking	1	1
System & Usage Logs	1	1
[Theme] Self-Reported & Elicited Trust Perception	3	3
Ambivalent Trust	1	1
Explicit User Feedback	1	1
Indirect explicit questions	1	1
[Theme] Trust Modeling & Measurement Frameworks	3	20
Automated Testing for Trust	1	1
Layered Trust Measurement	1	19
T1 - Baseline Digital Trust	1	1
T10- Explainability and Transparency Score	1	1
T11- Feedback Correction Loop	1	1
T12 - Personal Trust Adaptation Rate	1	1
T13 - Consensus Confidence Level	1	1
T14 - Meaning Alignment Score	1	2
T15 - Personalization and Privacy Paradox	1	1
T2 - Individual Psychometric Profile	1	1
T3 - Sovereignty Alignment Score	1	1
T4 - Dyadic Profile Fit	1	1
T5 - Consonance vs Dissonance Index	1	3
T6 - Boundary Violation Signal	1	1
T7- Comfort or Acceptance Score	1	1
T8 - Outcome Accuracy Metric	1	1
T9 - Contextual Fit Score	1	1
PDR Scorecard	1	1

Table C-5. Frequency of themes and approaches used to measure and monitor user trust in AI systems (Q7).

Name	Sources	References
[Q8] User Base Characteristics and Usage Scale	12	16
[Theme] Usage Scale	4	4
Large-scale public, research platform	1	1
Mid-scale deployment	1	1
Specialized national clinical network	1	1
Small-scale organizational use	1	1
[Theme] User Persona & Characteristics	12	12
Academics, Researchers	1	1
AI Experts, Data Engineers, Data Scientists	1	1
Clinicians (General)	1	1
Healthcare Administrators, Billing Staff	1	1
HR Professionals	1	1
Neuroscience & Medical Researchers	1	1
Patients	1	1
Skilled Knowledge Workers	1	1
System-as-User	1	1
Automated agents	1	1
System B as the main user of AI System	1	1
Young Students (17–25)	1	1

Table C-6. Frequency of user base characteristics and deployment scales discussed by participants (Q8).

Name	Sources	References
[Q9] User Trust Challenges in AI Systems	6	27
[Theme] Data Privacy, Security & Ownership Concerns	4	6
Access to my private and PII data	3	4
Mixing different customer data (competitive risk)	1	1
Use of personal data for retraining	1	1
[Theme] Governance, Accountability & Validation Gaps	2	3
Lack of trust in cross-institutional AIS Validation in Healthcare	1	1
System validation challenges (context and circumstances dependent)	1	1
Unclear accountability	1	1
[Theme] Human Factors, Perception & Behavioral Barriers	2	3
Anthropomorphizing the system	1	1
Experts distrust AI (overconfidence in own judgment)	1	1
Negative past experiences	1	1
[Theme] Performance, Reliability & Output Quality	3	7
Delayed response to issues caused by AI	1	1
Inaccurate or Misleading result	2	2
Low quality outputs	1	2
System Hallucination	2	2
[Theme] Personalization & Context Misalignment	1	1
Reduced usefulness due to lack of personalization	1	1
[Theme] Societal & Organizational Impact Concerns	2	2
Job replacement concerns	1	1
Shift from patient-centered communication → AI-oriented communication.	1	1
[Theme] Transparency, Explainability & Interpretability Issues	2	4
Black-box AI and lack of explainability	1	1
Misleading Explainability	1	2
Transparency used for wrong purpose	1	1

Table C-7. Frequency of themes related to user trust challenges in AI systems (Q9).

Name	Sources	References
[Q10] Features Enhancing User Trust and Confidence	13	30
[Theme] Data Privacy & Control	1	1
AI's limited access to user's PII data	1	1
[Theme] Governance, Accountability & Ethical Design	4	4
Accountability and Auditability	1	1
Clear ownership	1	1
Fair system	1	1
RAI integrated into the AI lifecycle, not as a supplemental layer	1	1
[Theme] Human Agency & Control	5	6
Human agency	3	3
Human in-the-loop	1	1
Let AI support, not replace	1	1
Opt-out option for users	1	1
[Theme] Perception, Trust Signals & Reputation	2	2
Perceived Positive Brand Reputation	2	2
[Theme] System Performance & Usefulness	6	9
Accuracy	1	1
AI helps with decision making	1	1
Convenience & Efficiency	1	1
Reliability	1	1
Responses grounded in accurate references	1	1
Usability	3	4
[Theme] Transparency, Explainability & Evidence	4	4
Clinical Evidence	1	1
Explainability	3	3
[Theme] User Empowerment & Engagement	2	3
AI Literacy	1	2
User Involvement in Problem Definition	1	1

Table C-8. Frequency of features identified as enhancing user trust and confidence in AI systems (Q10).

Appendix D: Figures Supporting the Thematic Analysis

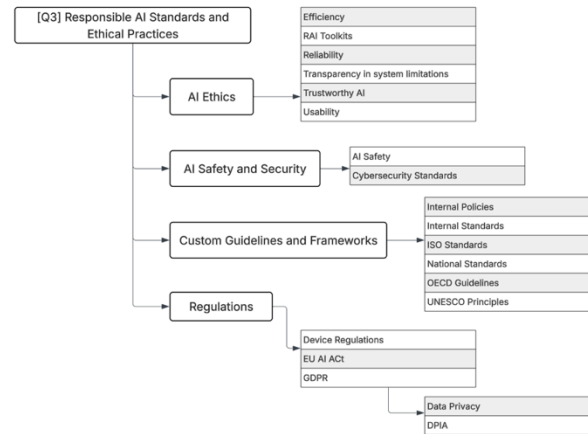


Figure D-1. Thematic structure of Responsible AI Standards and Ethical Practices (Q3).

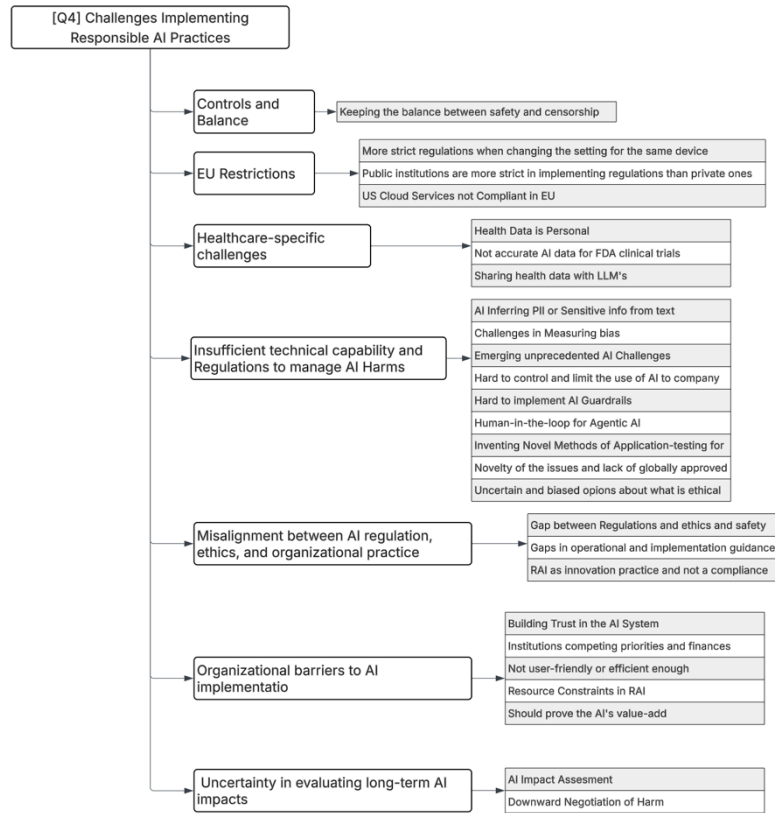


Figure D-2. Challenges in implementing responsible AI practices (Q4).

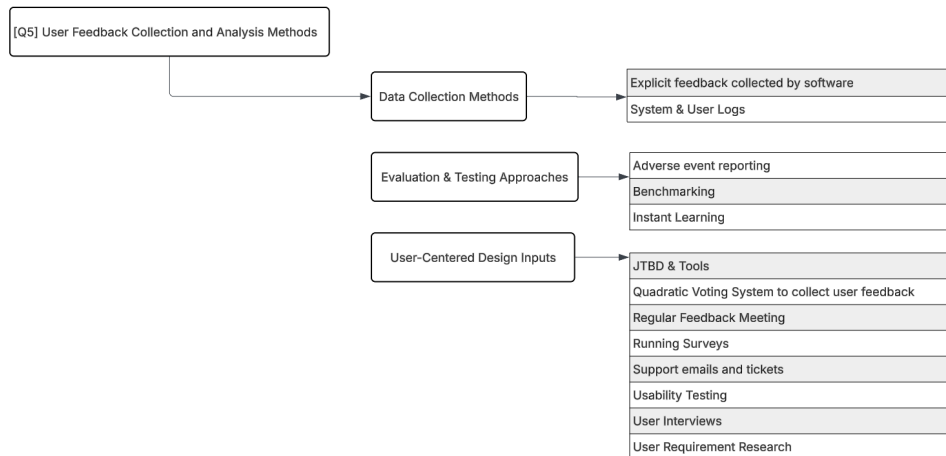


Figure D-3. User feedback collection and analysis methods used in AI system evaluation.

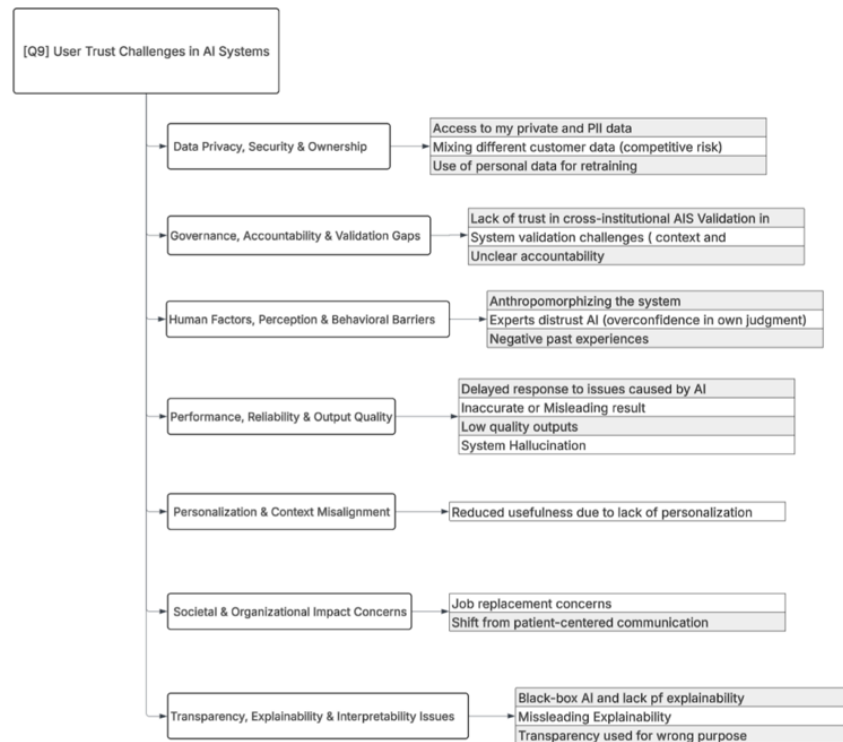


Figure D-4. Thematic structure of user trust challenges in AI systems (Q9), showing the technical, organizational, human, and governance-related factors that influence trust.